

# Erdős Problem #302

EP-302

L1 - TRACTABLE

OPEN

Number Theory

## ORIGINAL PROBLEM STATEMENT

Let  $f(N)$  be the size of the largest  $A \subseteq \{1, \dots, N\}$  such that there are no solutions to  $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$  with distinct  $a, b, c \in A$ ?

Estimate  $f(N)$ . In particular, is  $f(N) = (\frac{1}{2} + o(1))N$ ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>